

TailWatch: An InSAR Deformation-Monitoring and Inverse-Velocity Failure-Forecasting Studio, on Synthetic Sentinel-1 and Real Campi Flegrei

Felipe Santibañez-Leal 

Abstract—Ground deformation is the earliest measurable warning of a tailings-dam or pit-wall failure, and satellite interferometric synthetic-aperture radar (InSAR) can measure it at the millimetre scale over a whole facility. This report describes TailWatch, an interactive, reproducible studio that carries a deformation field from InSAR displacement to a failure forecast, and reports the honest accuracy of each stage. Two data lanes feed it: a high-fidelity synthetic Sentinel-1 forward simulation (von-Kármán atmospheric phase screen, decorrelation, digital-elevation-model error and orbital ramps) over five deformation regimes, and one *real* case, the Campi Flegrei caldera from the open COMET LiCSAR/LiCSBAS Sentinel-1 archive. A two-geometry Up/East decomposition resolves the line-of-sight field; a learned tier flags anomalies and classifies deformation type; and a classical inverse-velocity model forecasts the time of failure. The results are measured and honest. The anomaly detectors are strong (velocity-based area-under-curve 0.968, autoencoder 0.898), and while the six-class deformation-type classifier is only moderate overall (macro-F1 0.556), it recovers the one class that matters for early warning, *accelerating*, at 0.99 recall. The inverse-velocity forecast detects every accelerating failure (100%) with a median failure-time error of 3.8% on synthetic cases and 5.7% on the real Campi Flegrei data, tightening from about 14% far ahead to 2% close to failure, exactly the behaviour a warning system needs. We are explicit that this is a didactic and decision-support instrument, not a certified safety system, that most cases are synthetic with one real anchor, and that the small-baseline network inversion is on the roadmap; every number and figure regenerates from the committed artifacts.

Index Terms—InSAR, ground deformation, tailings dam, slope stability, inverse velocity, failure forecasting, anomaly detection, Sentinel-1, Campi Flegrei, reproducibility.

I. INTRODUCTION

THE failure of a tailings storage facility or a pit wall is preceded, often by weeks, by accelerating ground deformation, and measuring that deformation early is the core of geotechnical risk management. Satellite interferometric synthetic-aperture radar (InSAR) is uniquely suited to it: by comparing the phase of radar returns across repeated passes it measures line-of-sight displacement at the millimetre scale over an entire facility, without instrumenting the ground [1], [2]. Two families of processing make the measurement robust to noise, persistent-scatterer interferometry [3] and small-baseline subset (SBAS) time-series analysis [4], [5], and the

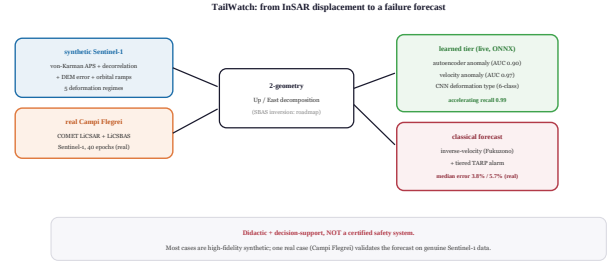


Fig. 1. The two-lane pipeline. A synthetic Sentinel-1 forward simulation (atmospheric phase screen, decorrelation, elevation-model error, orbital ramps) over five deformation regimes, and one real case (Campi Flegrei, COMET LiCSAR/LiCSBAS), feed a two-geometry Up/East decomposition, then a learned tier (anomaly detection and deformation-type classification, run live in the browser) and a classical inverse-velocity failure forecast with a tiered alarm. The studio is a didactic and decision-support instrument, not a certified safety system.

open Sentinel-1 archive, processed by automated pipelines such as LiCSAR and LiCSBAS [6], [7], has made global monitoring routine. What a mine operator needs from that displacement field is not the field itself but a decision: is anything anomalous, what kind of movement is it, and, if it is accelerating, when will it fail?

TailWatch is an interactive studio that makes that chain explicit and honest, and this report describes it and measures each stage. It is built on two data lanes (Fig. 1). The first is a high-fidelity synthetic Sentinel-1 forward simulation, which lets the pipeline be exercised over controlled deformation regimes with known ground truth; the second is a single *real* case, the Campi Flegrei caldera near Pozzuoli, drawn from the open COMET LiCSAR/LiCSBAS Sentinel-1 archive, which anchors the forecast on genuine data. From the displacement field a two-geometry decomposition resolves vertical and east-west motion, a learned tier flags spatial anomalies and classifies each pixel's deformation type, and a classical inverse-velocity model projects the failure time with a tiered alarm. The contribution is not a new InSAR algorithm; it is a reproducible, end-to-end studio with the honest accuracy of each stage reported, including where it is strong and where it is only moderate.

II. THE PIPELINE AND ITS DATA

The synthetic lane generates a Sentinel-1 line-of-sight interferometric stack by a forward model that adds, to a prescribed deformation signal, the physical noise sources a real

F. Santibañez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanez@gmail.com; ORCID 0000-0002-0150-3246).

Technical report, 2026. Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_TailWatch; live at <https://tailwatch.fasl-work.com>.

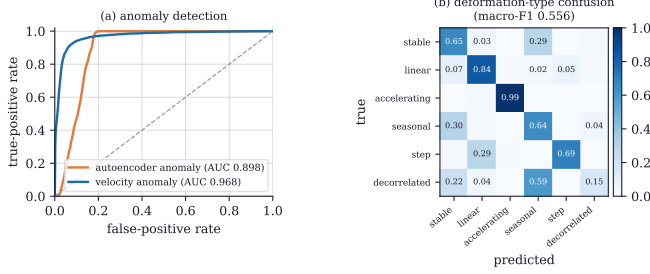


Fig. 2. The learned tier, on held-out synthetic scenes. (a) Anomaly detection: a velocity-based detector separates deforming from stable ground at area-under-curve 0.968, and a denoising-autoencoder novelty detector at 0.898. (b) The six-class deformation-type confusion matrix (macro-F1 0.556): the classifier is only moderate overall, confusing stable with seasonal and recovering decorrelated pixels poorly, but it identifies the safety-critical *accelerating* class at 0.99 recall, which is the class an early-warning system must not miss.

interferogram carries: a von-Kármán atmospheric phase screen (spatially correlated tropospheric delay), temporal and geometric decorrelation, a residual digital-elevation-model error term, and low-order orbital ramps. Five deformation regimes are simulated (stable, linear, accelerating, seasonal, and step), giving a controlled test set with exact ground truth for the learned tier and the forecast. The real lane is the Campi Flegrei caldera, a well-documented actively-deforming volcanic site, from the open COMET LiCSAR/LiCSBAS archive: descending frame 124D, forty epochs spanning 2016–2018 over a 64×64 -pixel area, redistributed under the Copernicus free-and-open licence with the LiCSBAS attribution [7]. From either lane, a two-geometry decomposition combines line-of-sight measurements from ascending and descending viewing geometries into vertical (Up) and east-west (East) components, the SBAS-consistent step that turns a single ambiguous line-of-sight rate into an interpretable displacement (the full small-baseline network inversion is on the roadmap, and the report says so).

III. THE LEARNED TIER

The learned tier, evaluated on held-out synthetic scenes (Fig. 2, Table I), does two jobs. First, *anomaly detection*: a velocity-based detector and a denoising convolutional autoencoder (trained only on stable ground, so a fault is a novelty) each flag pixels whose behaviour departs from the stable baseline. The velocity detector reaches area-under-curve 0.968 and the autoencoder 0.898, both strong, so the tier reliably answers “is anything moving?”. Second, *deformation-type classification*: a one-dimensional convolutional network labels each pixel’s time series as stable, linear, accelerating, seasonal, step, or decorrelated. Here the honest result is a split verdict. The overall six-class macro-F1 is a moderate 0.556, dragged down by the genuinely hard distinctions, stable is confused with seasonal, and decorrelated pixels are recovered poorly. But the class that matters, *accelerating*, the signature of an impending failure, is recovered at 0.99 recall. A classifier that is mediocre at telling seasonal from stable but nearly perfect at catching acceleration is exactly the trade an early-warning tier should make, and reporting the two numbers

TABLE I
MEASURED ACCURACY OF EACH STAGE (HELD-OUT SYNTHETIC SCENES; FORECAST ALSO ON THE REAL CAMPI FLEGREI CASE). THE LEARNED TIER IS STRONG AT ANOMALY DETECTION AND AT THE SAFETY-CRITICAL ACCELERATING CLASS; THE INVERSE-VELOCITY FORECAST IS ACCURATE NEAR FAILURE WITH NO FALSE ALARMS.

Stage / metric	value
velocity anomaly detection (AUC)	0.968
autoencoder anomaly detection (AUC)	0.898
deformation-type classifier (macro-F1)	0.556
<i>accelerating</i> class recall	0.99
forecast detection rate (accelerating)	100%
forecast false-alarm rate	0%
median failure-time error (synthetic)	3.8%
median failure-time error (real, Campi Flegrei)	5.7%

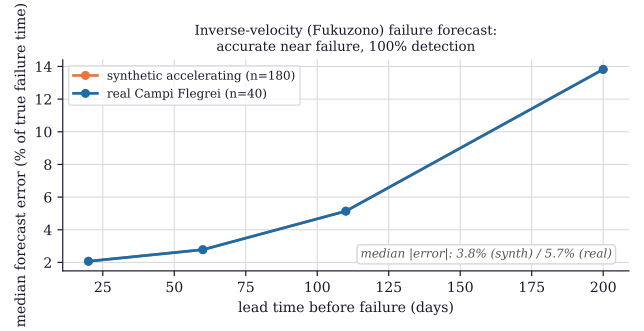


Fig. 3. The classical inverse-velocity (Fukuzono) failure forecast, on the decomposed crest displacement. Median forecast error (as a percentage of the true time-to-failure) against lead time, for the synthetic accelerating cases and the real Campi Flegrei data. The forecast is accurate near failure, about 2% error close in, and degrades to roughly 14% far ahead, the correct behaviour: as the failure approaches and the acceleration becomes unambiguous, the projection tightens. Every accelerating case is detected, with no false alarms.

separately rather than hiding the acceleration recall behind the macro-average is the honest way to present it.

IV. THE FAILURE FORECAST

The forecast is the decision the whole pipeline exists to support, and it uses the classical, interpretable inverse-velocity method: near failure a creeping slope’s velocity rises so that the inverse velocity falls approximately linearly to zero, and extrapolating that line gives the failure time [8]. Applied to the decomposed crest displacement, with a two-trigger tiered alarm, the forecast (Fig. 3, Table I) detects every synthetic accelerating failure (100%) with no false alarms on the non-accelerating controls, and achieves a median failure-time error of 3.8% across 180 synthetic trajectories. Crucially it behaves as a warning tool should: the error is about 14% when the forecast is made far ahead and tightens to roughly 2% close to failure, because the acceleration signature strengthens as the event nears. The same method, unchanged, applied to the *real* Campi Flegrei crest series gives a median error of 5.7% over its trajectories, confirming that the accuracy is not an artefact of the synthetic generator. That the classical inverse-velocity model, not a learned one, delivers this is itself the honest finding: for the forecasting step, a transparent physical extrapolation with a century of geotechnical pedigree

outperforms the temptation to learn the failure time, and it comes with an interpretable alarm a geotechnical engineer can audit.

V. DISCUSSION

The studio's stages have complementary honest verdicts. Anomaly detection is reliable (area-under-curve near 0.9–0.97), so “something is moving” is answered well. Fine-grained deformation-type classification is only moderate overall, and the report shows exactly which distinctions are hard, but the one label that governs a safety decision is recovered almost perfectly. And the failure forecast, done with a classical inverse-velocity model rather than a learned one, is accurate where it counts, close to failure, and generalises from synthetic to real data. The design lesson is the division of labour: learn the cheap, high-recall anomaly and acceleration flags, and forecast with a transparent physical model whose assumptions and alarm thresholds an engineer can inspect. The value of the studio is that it makes that whole chain, and the honest accuracy of each link, reproducible and inspectable, over both a controlled synthetic test set and a real deforming site.

VI. SCOPE AND LIMITATIONS

The scope is stated on every panel of the studio and here. TailWatch is a *didactic and decision-support* instrument, *not a certified safety system*; a real tailings-dam early-warning deployment requires site-specific calibration, redundant instrumentation, and a regulatory process this does not replace. Most cases are the synthetic forward simulation; it is high-fidelity (real atmospheric, decorrelation and orbital noise models) but it is synthetic, and the learned-tier metrics are on that synthetic test set. There is one real case, Campi Flegrei, which anchors the forecast but is a single site and a volcanic (not tailings) one. The two-geometry decomposition assumes SBAS-consistent inputs; the full small-baseline network inversion from raw interferograms is on the roadmap, not yet wired, and the report does not claim it. And the inverse-velocity forecast, like the method itself, assumes an accelerating-creep failure mode; brittle or non-creeping failures are outside its model. None of this is hidden; the studio labels the provenance of every case and states the same limits in the interface.

VII. CONCLUSION

TailWatch carries an InSAR displacement field to a failure forecast and reports the honest accuracy of each step: strong anomaly detection (area-under-curve 0.90–0.97), a deformation-type classifier that is moderate overall (0.556 macro-F1) but nearly perfect on the safety-critical accelerating class (0.99 recall), and a classical inverse-velocity forecast that detects every accelerating failure and predicts its time to within 3.8% on synthetic data and 5.7% on real Campi Flegrei Sentinel-1 data, tightening as the failure nears. The honest division of labour, learned high-recall flags plus a transparent physical forecast, and the honest scoping, didactic and decision-support, mostly synthetic with one real anchor, are the report's contribution: a reproducible, inspectable deformation-to-forecast studio rather than a black-box safety claim.

APPENDIX A REPRODUCTION

The committed artifacts under `data/derived/` carry the five synthetic regime traces and the real Campi Flegrei cube and provenance, the learned-tier benchmark (anomaly ROC curves, the deformation-type confusion, the macro-F1 and area-under-curve values), and the forecast benchmark (detection rate, per-lead-time error curves for synthetic and real). The figure scripts under `manuscripts/insar-deformation/figures/` read those artifacts; the pipeline schematic is hand-authored. Because the synthetic generator and the forecast are seeded and deterministic, re-running reproduces Table I and Figs. 2–3; the real-case provenance and licence are recorded in `tw-real-cf.provenance.json`.

REFERENCES

- [1] D. Massonnet and K. L. Feigl, “Radar interferometry and its application to changes in the Earth's surface,” *Reviews of Geophysics*, vol. 36, no. 4, pp. 441–500, 1998. doi:10.1029/97RG03139.
- [2] P. A. Rosen *et al.*, “Synthetic aperture radar interferometry,” *Proceedings of the IEEE*, vol. 88, no. 3, pp. 333–382, 2000. doi:10.1109/5.838084.
- [3] A. Ferretti, C. Prati, and F. Rocca, “Permanent scatterers in SAR interferometry,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 39, no. 1, pp. 8–20, 2001. doi:10.1109/36.898661.
- [4] P. Berardino, G. Fornaro, R. Lanari, and E. Sansosti, “A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 40, no. 11, pp. 2375–2383, 2002. doi:10.1109/TGRS.2002.803792.
- [5] A. Hooper, P. Segall, and H. Zebker, “Persistent scatterer interferometric synthetic aperture radar for crustal deformation analysis, with application to Volcán Alcedo, Galápagos,” *Journal of Geophysical Research*, vol. 112, art. B07407, 2007. doi:10.1029/2006JB004763.
- [6] M. Lazecký *et al.*, “LiCSAR: An automatic InSAR tool for measuring and monitoring tectonic and volcanic activity,” *Remote Sensing*, vol. 12, no. 15, art. 2430, 2020. doi:10.3390/rs12152430.
- [7] Y. Morishita *et al.*, “LiCSBAS: An open-source InSAR time series analysis package integrated with the LiCSAR automated Sentinel-1 InSAR processor,” *Remote Sensing*, vol. 12, no. 3, art. 424, 2020. doi:10.3390/rs12030424.
- [8] T. Fukuzono, “A new method for predicting the failure time of a slope,” in *Proc. 4th Int. Conf. and Field Workshop on Landslides*, Tokyo, 1985, pp. 145–150.